

APPM 5120 - Graduate Project Proposal

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Background/Motivation

In discussions with a small local electrical contracting company, they expressed a need for a more effective system to help determine when they should to bid on new projects. The company wants to minimize unused core (in house) labor while maximizing expected profits under various labor, project start dates, project durations, and external hiring constraints. The timeline of a given project is as follows: First, a job is announced and a deadline is set for when companies need to submit their bid. After the deadline passes, the company is either awarded the job or not. If they are awarded the job, they are expected to start work within the provided acceptable start date window.

One key aspect of this project is the element of time. Rather than solving the problem for a single timestep, I propose creating a simulation using a rolling optimization strategy to determine optimal bidding decisions at each timestep over the course of a predefined timestep window. To do this, I plan to use synthetic data to simulate scenarios of applications becoming available, deadlines coming and going, and bids being accepted and denied. For example, a bid being accepted would lock in the labor the company proposed to allocate for the given job whereas the bid being denied would free up that labor to be allocated for other bids. For this project I'm proposing formulating and implementing a simplified version of the problems this company is facing and, time permitting, will extend the project to include additional implementation details the company is interested in. Generally speaking, the result of the algorithm will provide a recommendation for which projects should be bid on (derived from $S_{p,t}$), the number of core workers assigned to project p at time t with skill s ($F_{p,t,s}$) and, finally, the number of non-core workers that will need to be hired on for project p at time t of skill s (derived from $F_{p,t,s}$ and $D_{p,t,s}^{\text{active}}$). Below is a brief formulation of the problem.

Parameters

There are a few external parameters which the algorithm will expect. The parameters $C_{t,s}$, $D_{p,\tau,s}$, H_p , R_p , A_p , and P_p come from company information and estimates. The parameters E_p and L_p are taken from the bid application.

$C_{t,s}$:	Number of available core workers of skill s at time t	(1)
$D_{p,\tau,s}$:	Estimated labor demand for project p at relative time τ with skill s	(2)
H_p :	Time length of project p	(3)
R_p :	Required core-to-total labor ratio for project p	(4)
A_p :	Estimated probability of bid acceptance for project p	(5)
P_p :	Estimated profit for project p	(6)
E_p :	Earliest feasible start time for project p	(7)
L_p :	Latest feasible start time for project p	(8)

Decision Variables

The output of the algorithm at a given timestep will include the following.

$F_{p,t,s}$: Number of core workers assigned to project p at time t with skill s (9)

$S_{p,t}$: Binary indicator for whether project p starts at time t (10)

Intermediate Definitions

$$D_{p,t,s}^{\text{active}} = \sum_{\tau=0}^{H_p-1} D_{p,\tau,s} S_{p,t-\tau} \quad \text{Active demand for project } p \text{ at time } t \text{ based on selected start time} \quad (11)$$

$$W_{t,s} = \sum_p D_{p,t,s}^{\text{active}} \quad \text{Total labor demand for skill } s \text{ at time } t \quad (12)$$

$$T_{p,t} = \sum_s D_{p,t,s}^{\text{active}} \quad \text{Total labor demand for project } p \text{ at time } t \quad (13)$$

$$U_{t,s} = C_{t,s} - \sum_p F_{p,t,s} \quad \text{Unused core labor of skill } s \text{ at time } t \quad (14)$$

Objective

Minimize unused core labor and maximize expected profit:

$$\min z = \sum_t \sum_s U_{t,s} - \sum_p \sum_{t=E_p}^{L_p} A_p P_p S_{p,t} \quad (15)$$

Constraints

1. Core worker capacity constraint

The number of core workers assigned cannot exceed the available capacity.

$$\sum_p F_{p,t,s} \leq C_{t,s} \quad \forall t \in T, s \in S \quad (16)$$

2. Core labor cannot exceed project demand

$$F_{p,t,s} \leq D_{p,t,s}^{\text{active}} \quad \forall p \in P, t \in T, s \in S \quad (17)$$

3. Core-to-total labor ratio constraint

A project must satisfy the required core labor ratio whenever it is active.

$$\sum_s F_{p,t,s} \geq R_p \sum_s D_{p,t,s}^{\text{active}} \quad \forall p \in P, t \in T \quad (18)$$

4. At most one start time per project

A project is bid on only if exactly one feasible start time is selected.

$$\sum_{t=E_p}^{L_p} S_{p,t} \leq 1 \quad \forall p \in P \quad (19)$$

5. Start time must lie within the feasible start window

$$S_{p,t} = 0 \quad \forall p \in P, t \in T \text{ such that } t < E_p \text{ or } t > L_p \quad (20)$$

6. Binary start decision

$$S_{p,t} \in \{0, 1\} \quad \forall p \in P, t = E_p, \dots, L_p \quad (21)$$

7. Nonnegativity of core labor assignment

$$F_{p,t,s} \geq 0 \quad \forall p \in P, t \in T, s \in S \quad (22)$$

Solution Assumptions

All constraints and the objective function are linear and the decision variables are integers so the problem is a Mixed-Integer Linear Program. The variable $S_{p,t}$ are binary variables and the variable $F_{pt,s}$ contains integers. I believe I'll need to look at Branch-and-Bound algorithms to try and solve this problem.